

## Data

### Source:

ESCO(European Skills, Competences, Qualifications and Occupations)

| Dimension       | ESCO( <a href="#">Use ESCO   European Skills, Competences, Qualifications and Occupations (ESCO)</a> )                                                                                                                                                    | Online Courses ( <a href="#">Online Courses</a> )                                                                                                                                                                                                                                      |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Volume</b>   | <b>Raw:</b> ISCOGroups en.csv (619 groups), occupationSkillRelations_en.csv (126,051 rows), and skills_en.csv (13,960 concepts).<br><b>Cleaned:</b> isco_clean.csv (619 groups), skills_clean.csv (2,154 concepts), and relations_clean.csv (9,376 rows). | <b>Raw:</b> 8,092 rows and 45 columns; Coursera (2,819), FutureLearn (4,843), Udacity (282), and Simplilearn (148).<br><br><b>Cleaned:</b> 5,275 rows and 24 columns; Coursera (2,819), FutureLearn (2,031), Udacity (277), and Simplilearn (148).                                     |
| <b>Velocity</b> | Static dataset downloaded once; there is no real-time intake.                                                                                                                                                                                             | Static dataset downloaded once; there is no real-time intake.                                                                                                                                                                                                                          |
| <b>Variety</b>  | Structured CSV files containing free-text labels, hierarchy codes, and URI keys that link four tables.                                                                                                                                                    | Course data collected from four platforms: Coursera, FutureLearn, Udacity, and Simplilearn.                                                                                                                                                                                            |
| <b>Veracity</b> | No duplicate rows were found in the CSV files. The 23 rows missing skillType in occupationSkillRelations en.csv were backfilled from skills clean.csv by tracing them to five specific skills.                                                            | Five duplicate Udacity rows were removed. A total of 4,843 scraped entries were consolidated into 2,031 unique FutureLearn courses. Some Coursera courses share a title and some Simplilearn courses share a URL; these records were retained because they refer to different courses. |

Online Courses, Kaggle: course catalog sourced from Coursera, FutureLearn, Udacity, SimpleLearn.

## Input, Output, Preprocessing

### Model

We combine two models, each solving one part of the task. Model 1 sorts a course into a subject so the system can narrow the catalogue; Model 2 matches a course's free-text skills to ESCO skills so the system can explain where a study path leads.

#### Model 1: Course → Category (supervised classification)

**Input:** course title + short intro. **Output:** one of 11 categories + confidence. **Data:** the 2,577 labelled Coursera courses left after removing duplicates; fixed stratified split of 1,805 train / 386 dev / 386 test (seed 42). Three models were compared on dev; the best dev macro-F1 was selected and the test set was scored only once.

| Model                                          | Dev accuracy | Dev macro-F1 | Training                            | Inference           |
|------------------------------------------------|--------------|--------------|-------------------------------------|---------------------|
| <b>Majority-class baseline</b>                 | 0.321        | 0.044        | –                                   | <0.01 ms/item       |
| <b>TF-IDF + Logistic Regression (selected)</b> | <b>0.806</b> | <b>0.752</b> | <b>0.62 s</b>                       | <b>0.13 ms/item</b> |
| <b>TF-IDF → SVD(300) → MLP (256 hidden)</b>    | 0.811        | 0.724        | 7.18 s, 92 epochs, final loss 0.022 | 0.09 ms/item        |

**Test result (Logistic Regression): accuracy 0.795, macro-F1 0.772.** Macro-F1 is reported next to accuracy because classes are very uneven (Business 32%, Math & Logic only 22 courses); always guessing Business already gives 32% accuracy. The weakest classes are the smallest (Social Sciences F1 0.58, Personal Development 0.64). The MLP's training loss drops almost to zero while its dev macro-F1 is lower, so it overfits the 1.8k training examples; we therefore kept the simpler linear model.

## Model 2: Skill → ESCO (semantic retrieval)

**Input:** one free-text course skill. **Output:** top-5 ESCO skills + similarity score. Nothing is trained on our data: ESCO skills are ranked by cosine similarity using a pretrained sentence encoder (all-MiniLM-L6-v2) or word + character TF-IDF. Exact string matching links only 7.4% of the 4,229 distinct course skills to ESCO, so a lookup table is not enough. With no labelled course-ESCO pairs, we held out one alternative label per ESCO skill and used it as a query with a known answer (index: 2,154 skills).

| Method                                 | Top-1        | Top-5        | MRR@5        |
|----------------------------------------|--------------|--------------|--------------|
| Exact string match (baseline)          | 0.000        | —            | —            |
| Word + character TF-IDF cosine         | 0.816        | 0.927        | 0.864        |
| Sentence embeddings (all-MiniLM-L6-v2) | <b>0.866</b> | <b>0.962</b> | <b>0.905</b> |

On the real course skills, 65.9% reach a similarity of at least 0.35 (TF-IDF).

## Model adequacy

Both are standard machine-learning methods sized to the data. A keyword lookup is too simple (matches only 7.4% of skills); fine-tuning or training a transformer on ~1.8k labelled texts would be too complex and would overfit. Everything runs on a CPU in seconds (Intel, Windows 11), no GPU needed.

## Task-Architecture Fit

Each architecture matches what its sub-task actually requires. Model 1 is a closed-set classification problem over a modest labelled set (1.8k training texts), so a linear model on sparse TF-IDF features is the right size: the dev comparison shows it already matches the higher-capacity MLP on accuracy while generalising better, so nothing is gained by adding depth. Model 2 has no labelled course-to-ESCO pairs to train on and must generalise to skill phrasings it has never seen, which rules out a supervised classifier; a pretrained sentence encoder used purely for retrieval fits that constraint without requiring any task-specific training data. Sentence embeddings naturally align with short phrase semantics without truncation issues, and pre-indexing vectors enables fast, scalable nearest-neighbor retrieval without slow cross-encoder overhead.

## Evaluation vs. Real-World Domain Shift

The 86.6% Top-1 accuracy is measured on formal ESCO synonyms, meaning real course skills—where 34.1% fall below a 0.35 similarity threshold—may face lower real-world precision and require confidence score cutoffs.

## Software

**Environment:** Python 3.13.9, Windows 11, CPU-only (no GPU); conda-managed (environment.yml), Git for version control, fixed seed 42.

**Data:** Loading, cleaning, deduplication, and descriptive statistics use pandas 2.3.3 and NumPy 2.3.5. Exploratory profiling charts and tables in the data-cleaning notebooks use matplotlib 3.10.6, seaborn 0.13.2 and IPython 9.7.0.

**Model:** scikit-learn 1.7.2 provides both models' machinery — TF-IDF, TruncatedSVD, LogisticRegression, MLPClassifier and the majority baseline for Model 1, and TF-IDF + cosine similarity plus all metrics (accuracy, macro-F1, per-class report, MRR) for evaluation. Model 2's neural retriever uses sentence-transformers 5.5.1 (all-MiniLM-

L6-v2, PyTorch 2.12.0 backend). Fitted pipelines are saved with joblib 1.5.2; the confusion matrix and training-loss curve use matplotlib.

**Reproducibility.** Notebooks are executed headlessly with Jupyter nbconvert 7.16.6; every reported number is written to data/processed/ by the notebook that produced it rather than transcribed from a cell, so the report is assembled from files.